

Kathryn Nave ~ Active Inference Insights 020 ~

Agency, Vision, Life

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hello everyone and welcome back to active inference insights as always I'm your host darus PvZ Wayne and today I'm really excited to be speaking with Kate Nave Kate is an early career research fellow at the University of Edinburgh and her research focuses on developing a realist account of autonomy and agency grounded in the uniquely metabolic existence of living systems she has also just finished writing a book titled a drive to survive the free energy principle and meaning of life which I know the whole active influence Community is looking forward to reading and hearing about Kate welcome to the show thank you so much for coming on it's going to be really fun I hope yeah thanks for having me nice to meet you so there were a couple books as you know about active inference but um they're all quite text Booky and Technical so I'm kind of curious about just generally how did the book idea come about uh and what was it like writing this book and you know overall sure yeah it was I guess quite a securious process it kind of began during my PhD um when I was looking at actually began with looking at predictive processing and visual phenomenology from like kind of a perspective in hilan phenomenology trying to connect that to predictive processing which was interesting but um through that I got more into the frery principle um and I think I initially began with thinking that the frery principle was going to be a very useful way to kind of scale up in active accounts of purpose and intentionality and then like get through the course of kind of investigating that it ended up being kind of I guess the book came out of this to do two things for me um the first was to try and get through the kind of philosophical ideas of the FR principle in a way that doesn't sort of shy away from the formalism but doesn't rely on the formalism if you see what I'm saying because I think it's at the point which I was writing it was well pretty well commun being pretty effectively communicated to people with a mathematical background that wasn't perhaps initially the case um but the kind of communication of the mathematics I think had gotten a lot clearer but it was still the case that if you didn't have any kind of grounding in that there wasn't anything that was accessible to

you I think in book form um so the first half of it is really trying to look into what is the what are the what are the formal aspects of the principle actually saying in a way that's explained non- mathematically and intuitively um not to say I mean you know to some people mathematics is intuitive but to some people it's not um and then the second part is I think through that kind of deconstruction of what is the mathematics actually telling you it started to become clearer to me that the kind of aspects that were supposed to account for purposiveness and agency weren't really there which I mean I think now is becoming now maybe doesn't seem that stranger thing to say I don't know I think there's been more papers and more discussion of this aspect at the time I would say there wasn't so that was part of it and the other part of it was sort of discovering a whole lot of really cool work on the physics of living systems that shows how much more complicated they are than we expect how difficult they are to pin down um how sort of pran and changing they are and I think all that stuff is really cool and I found that the fu principal is quite a good launch pad for getting into that work because it sort of of pushes how far can we go with this sort of mathematical formulism how far can we get in our description of leing systems and for me like butting up against those limits introduced me to a whole load of ideas in kind of organicist philosophy of biology um in particular that is that's kind of where the book ends I suppose with a here's what's kind of Beyond The Finity principle here's all this stuff that isn't I don't think for me not something that can be a part of this like very nice very simple principle um and yeah so like kind of like a simple expl as simple as possible explanation of the constructs involved in the fry principle how far can they take us and what can't they help us with when it comes to living systems in particular sure see I I want there's there's loads of things there um I wanted to pick up on this thing of you mentioned agency and kind of Theology and the changing perception that we have of those Notions at least within Act of inference Act of inference might intuitively lend itself to a kind of realist notion of agency um just sort of embedded within the idea of self-evidencing in some way and this goes you know all the way back to kind of inactive Notions of autop poesis that the uh and maybe even earlier to hiigaran phenomenology that the organism is taking a stance on its being and that is self- evidencing that is it's carving out its own Niche and its own world um when you say that it's sort of not there do you mean that it's there are two kind of ways of interpreting that one is that it's actually not there on logically speaking that it's a nice illusion that we you know create for ourselves to get by in the world that's one stance then the other thing is that it can't really be fleshed out in terms of the formalisms and there might be something latent uh in the ontology which the epistemology

can't get us to at least the epistemology of active inference so which one is it is it the uh the instrumental tool of active inference is insufficient to give us agency and ontology or is it uh that it's kind of an illusion the whole project in the first place yeah I think it's very much that the the tool that active inference is doesn't allow us to describe the feature of reality that underpins this kind of purposiveness is how I see it um and I think and I can go into more of like what those features are but I think that's why you now have this kind of perhaps move in active inference to where previously self-evidencing um you know self models all of this was presented as very purposive I think I'm you're starting to kind of see more papers that are now saying that that purposive gloss on active inference is instrumental yeah that it's useful to describe certain systems as purposive and not so useful in the case of other systems but in all cases all that's happening is for energy minimization and perhaps a sort of datian kind of picture where when you get a certain level of complexity um it's helpful to use this kind of intentional gloss but it's just a gloss um and it's a gloss that picks up a real difference but it's um a difference of degree not a difference in kind I think is the important thing there so the question of where we draw the boundary on that difference of degree is relative to our interests um and that's not the view I have but I think that's the view that you get to if your ontology is limited to what the frery principle gives you because it doesn't give you these kind of differences in kind between what I would think of as agential and purposive systems I living systems and systems that aren't identical and purposive for me there's a difference in kind there um so does this rest on some kind of axiomatic uh presupposition that there is agency and purposiveness embedded within living systems and and where is where's that coming from is it just phenomenology um where where do we how can we ground this kind of starting point that well there is a difference really between me and and a rock let's say yeah I think that's the part of it um so that's not something I get into as much as I'd like to in the book that's kind of what I'm working work on now I think I began with that presupposition because I began with this inactive view of purposiveness and agency and I suppose because I began with working in visual experience I think the way that visual experience is characterized in a lot of the phenomenological accounts that I would draw upon is very purposive and intentional right it's an attempt to there's always talk physically there's very sort of striving notion of perception where it's about kind of bringing things into alignment and achievement and that kind of a very int purposeful gloss to it so I definitely began with that very like if you can't everything we want to say about cognition for me is grounded in agency um and if we get rid of that then the kind of whole framework does sort of crumble and I think you know people do feel that

you can keep it going with this um instrumentalist stance I think there's a lot of problems with that because I mean one of them is when we're talking about talk of agency purpose is being relative to like our interests and we already we've got interests in the background there right that we need to make sense of you know is it a good idea to speak if this is purposive or not it's relative to what I'm interested in how do I explain what I'm interested in you get that kind of regressive perspective which is normally stopped with Evolution I guess it's the standard move yeah um you go in natural selection um so there's a reason for thinking if we don't have agency I think we lose a lot more than we realize and there's this gorgeous photo actually like I love jry photo and he's obviously he's he's very known for his defense of intentionality and purposiveness in a very different way from which I'd want to defend it but he's like if that's gone then it's basically just the end of the world he just throws up his hands and cries um and I think they I do think there's something right about that like I think we lose more by letting go of that it's we don't just lose I'm not even that interested in Free Will it's not just that we free will we lose Notions like function we lose the ability to sort of demarcate systems relative to they do it all just becomes relative to something we don't really understand in terms of section so does the deeper question does the deeper Free Will question really uh feed into your consideration of agency I mean you said there that you're not too fuss about it in some ways it's a worthwhile thing not to be fussed about because it's a it's a philosophical wormhole in and of itself um does that fodorian quote and and your perspective rest or is it contingent in any way on there being free will or is it just kind of this uh parent tellos or this parent function that things have yeah so I think it's very it rests very much on the function aspect not being a parent okay when it being the case that it's not up to you or me what a particular living system is trying to do like if I say any coli is trying to metabolize glucose that is I want to argue um something that's true about that EO like independently of what you or I think about it and I'm quite an anti-realist um so for me that's quite a big move to say there are real facts in the world independent of our perspective and I think that's about the most basic one is these kind of facts about what living systems need those are like going to be yeah the most sort of basic realist facts that you get in your analogy um but the reason I say I'm not that interested in the Free Will aspect is I want to be able to say that certain systems have needs that they are trying to do things that they have things can be better or worse for them so it's about that kind of there is a sense of responsibility there where we can attribute these things to these systems it's just that the concept of free will I think is so it's one of those Concepts that's almost like a bit um contaminated in that it's very hard to agree what different people mean by it and it plays very

different roles in very different discussions so it's not when I say I'm not interested in it I think I'm interested in things that some people might consider as being what free will is about I just don't like to talk about it in terms of will because it's just too messy a concept yeah no I think I think it's absolutely right that we need to just at least be precise about what we mean by the words we use I mean it it ends up becoming a sort of you know people um just just spending half of their papers writing what can you know the meaning of the concepts they're using but a shared language game would be nice because I think um when people hear agency they do think of Free Will and they don't think of this uh this kind of non Lo like for me just this notion that agency is a function performative function uh it's kind of like I don't know for me it doesn't really feel like it's enough yeah but um I also wonder whether if we so we we take function and process as realist real in our ontology do we take uh do you take thingness to be real as well because this is a kind of contentious Point within an active inference well maybe not contentious but at least a point of curiosity which is active inferentially on the fact that there is a thing and that thing persists you know you can describe a thing as persisting such that it's the same thing today and tomorrow is that a viable or reasonable Axiom or do you think that that can be sort of demystified deboned yeah just come back to the thing you said first off about this not being sufficient for agency I would also the kind of thing I'm talking about I would also agree with that I tend to kind of refer to use the term autonomy for that reason because I think autonomy is like a sort of necessary condition of agency but maybe not the full-fledged thing and actually before we go on to the second Point by autonomy are you talking here about a sort of traditional and activist account of autonomy well I think where I would like Grant to originate maybe the notion of autonomy I'm interested in is just this kind of Canan thing of like being a system that gives a law to itself yeah so that's that plays out in an active concept of autonomy but I think it's a sort of more thinner or essential notion that people have developed in these different ways but the basic thing is how can you be a system that constrains yourself because you know if you're the only constraint on yourself and surely you can just choose to stop constraining yourself um like it's a really to me it's a really really lovely philosophical kind of um I guess an antimony in C or like a paradox where there's something really important about the idea of self-constraint it seems like a self-defeating idea and that's maybe that's kind of what I mean by autonomy is like this really all of the confusion about agency and all of the reasons that agency in general I think is difficult to explain is because it's for me I think it's Central ingredient is this kind of autonomy of being a system who has things that you have to do

because of the system that you are that aren't externally imposed so it is a very active Nation as well they're not internally imposed needs but leads you still have to satisfy them so what so what what kind of example could you give there so for examp I think obviously that se that additional Clause is necessary because then I can say well the rock is autonomous because it has needs at least uh you know from the mathematical statistical instrumental standpoint I.E to minimize variational free energy what needs could you say uh are do we impose on ourselves which are not I mean I'm I to put lay my cards on the table I'm personally quite quite skeptical of the internal external boundary in any case I'm not sure what it means to have a internal versus external constraints because underlying all of that you need to have some locus of free energy minimization right so yob har say it's the dorsal horn or you know Andy Clark might say it's me and my phone and so on but in any case uh I'm I'm quite happy to have a kind of process ontology about where those boundaries lie um but what is it so what would it mean I mean if we can say the rock clearly doesn't have kind of this interiority whereby it's uh creating its own norms and it's abiding by those Norms that I seem to be um can you give us an example of like one of those because for me I feel like I can always reduce it back to some kind of exteriority uh sure like so the way I think about this kind of self law is is then does become a more an activistic notion though so that this the kind of framework I particularly like for it is um the kind of biological autonomy framework that's been developed by um alv Marino mattio mosio in terms of constraint closure which is very inactivist but I wouldn't necessarily say they're inactivist so in that sense um but the the way the way that you have a system that has something like inter an interiority exteriority distinction but not obviously self what's it self like not self-standing completely where you have that kind of operation that kind of constraint closure which is their development of what would be organizational or operational closure in an activism but you know you have all these different versions of closure that are all actually really quite different um but sorry just a second to come back to it so the idea is the way in which that gives you this kind of need that is your own need that a rock doesn't have is I think in two ways firstly there's the sense in which I think when we speak of non-living things those are things those are carvings out of the world that are relative to our interests so something like a rock for me is not a natural kind um anything that's not living arguably I would say those are kind of distinctions that we're placing on the world we're saying to be a rock you need to not be heated over a certain temperature because then you will collapse and no longer be a rock but that's not really got anything to do with any sort of self-individuating system um that's just my definition of what a rock is so to say that you know you

have you can have this sort of statistical model of a rock and then you can say that for energy must be minimized relative to that model if it's to stay as a rock that's still very much like my imposition on the world when you have something that's I guess a metabolic system I think what you have is a system where there are needs intrinsic to that Network where each part of it depends depends on other parts in such a way that it's not about how to put this how this so that there's a stability but the stability depends upon the activity of the system so if you just leave it on its own um it will if you I always find this really difficult to talk about if you it's kind of like the it's kind of like I think the the classic example is something like the cell yeah then the cell has this uh partially permeable membrane and it needs the sorts of metabolic resources in which to continue to generate that partially permeable membrane I mean I think that's a classic autopoietic notion the thing is I'm being really annoying but why not it's philosophy um I just wonder whether that kind of so that thing of uh in terms of operational closure the individual constituent Parts rely on one another so without one you don't have the functioning of the the whole and and I think you know they term that precariousness um I just I wonder whether you could consider the in if there is the interiority of a rock to also be precarious in that way such that if the central bit of a rock is heated to a certain degree such that it melts the whole thing melts right in some sense the whole thing collapses and become soil or at least not the singular rock that we thought it was it would become a multiplicity of rocks I it's interesting that you carve out the sort of natural kind and and the rock is not necessarily A distinct thing in and of itself and it's an imposition I kind of in maybe it's a very uh exaggerated Buddhist idea that I think we're the same right I think that I think it's uh it's difficult to say that I am ultimately physically distinct from the chair on which I'm sitting now I think the one I I spoken to Carl about this and I think the the what he rests on is the idea of action uh and I guess what in an activism would be adaptivity so there kind of Ezekiel dealo idea that by because we can take a stance our own being we can change the world in accordance to our prior so to use active inference language how does that solution stand for you I mean I I you know just to sort of tee you off is there an issue with the underlying notion of action as well is that is that a Troublesome concept yeah um so I think it's funny like when you have this discussion about whether or not living system is a a self- individuating system or not that you you're either often talking to somebody who would just absolutely accept that rocks and chairs and things are just real things in the world independently of us or you're talking to somebody from your position on the Other Extreme of it and I find I suppose I find the the end of the spectrum that you're more sympathetic that's where I would have initially

gone lent before I was more convinced by the this idea of self- individuation and I think yeah so I think action is very pivotal but I think it's often used in a very weak or general way in which I think we can make a distinction between an action and a movement and if you just refer to all movements as actions then you're not going to get any sort of distinction between a movement operationally closed system and a rock because you can say that a rock depends upon all the forces that all of its different particles exert on each other to hold it together you can call those inter molecular forces actions right can say there's a network of actions between the various components of a rock keeping that rock going yeah yeah yeah so yeah this is helpful this is kind of what I was trying to get to before was that why is an operationally closed Network different from a system where the Integrity of the whole does depend on interactions between its parts we can think of a rock that way and I think it rests on having this notion a very I guess quite unusual but quite stringent notion of like what it is for something to be acting on its part where an action and this is what you get in the kind of constraint closure formulation is an action is when you're constraining a release of energy to perform work in a way which is General again like um you know a channel dug into the mountain channeling water down a hill is constraining the release of energy to perform work of turning a water wheel so the idea of constraining a release of energy to perform work is not sufficient but it becomes an action where when the constraint itself is dependent upon the work that it does when you have that kind of circularity so and that's what you do have in metabolism of course because in metabolism you have these enzymes that are catalyzing reactions that only happen because of the enzymes but if the reactions don't happen the enzymes themselves aren't regenerated and that's a very different relationship between Parts than you have in something like a rock and that's where I think you do start to get this more genuinely self- individuating thing and it's why I'm not super key on this kind of way in which precariousness is often talked about in the inactive literature because precariousness can sound very vague you know everything is precarious over time everything will collapse over some period of time it has to be a very particular kind of precariousness that is distinct to living systems which is I think this very particular kind of closure great no that's that's really useful um you mentioned constraints there a couple of times yeah um and I feel like there's a this is not a literature I'm super familiar with I know work of someone like alicio jarero and this notion this kind of Distinction Aristotelian distinction I think between causality and constraint in your own words what what do you understand as the difference between those is there again is this a natural you know a natural difference of natural kinds that there is something genuinely ontologically

different in the universe between causality and constraints uh are they the same thing with different you know words attached to them um what how do you understand them and also what does that distinction give us in terms of our scientific or philosophical Outlook yeah so I think one of the caveats here is I'm I've not done a whole load on the metaphysics of causation as it's kind of generally understood and I think what I love about constraint is constraints make sense to me when causation is talked about in a lot of the philosophical literature at least my sort of surface level of awareness of it is it gets very messy and you end up with views on which there's no such thing as causality because if you have a particular notion of causality you don't find that at the level of fundamental physics um and then you have I think among sort of philosophers maybe more like myself who are in something like cognitive science or not in the kind of philosophy of physics but in more General philosophy of science areas you can sometimes have like a very sort of traditional basic notion of like billiard ball causation an event happens it triggers another event and that's just not very useful for a lot of the kind of things you to talk about um whereas I think constraints are particularly for the kind of causation so I think constraints are a type of cause and for me they're the type of cause that makes the most sense um and they are the type of cause that helps you talk about living systems really well because they get that element of the way in which it's very difficult to talk about interiority and exteriority in the living system because there's such a dependence on exteriority there is also some sort of closure and what con the idea of constraint causation gets across is that dependence on exteriority because you need the you know you need the input of energy to be channeled by the constraint so it depends on something from the outside to do anything but the thing itself is what makes things happen and it gives you that I think also that sense of sorry uh being able to identify a cause because if you just have the billiard ball thing then you just Trace back that chain of events at the beginning of the universe and it's who started it whereas if you have a constraint plus even just like a random um spontaneous reaction you have two things that you don't need to go back on a causal chain like the event of a reaction happening only happened because this particular constraint was here and this spontaneous release of energy occurred so you have that sense in which it's much easier to identify I think it's much easier to pin down a cause when you think of it in terms of constrained releases of energy rather than events triggering other events although I can imagine a very picity philosophy of you know philosopher of science would say that you could also trace the constraint ad infinitum right or or add beginning of the universe because uh presumably you you know constraints presumably have some kind of like a fractal nature

such that something is constraining only because it was you know the result of some other constraint which in of itself was a result of some other constraint uh I mean that's just addendum I think I think they're both useful ideas um I don't know my experience is that we're as you say stuck very much still in the kind of humean um skepticism which I think is totally reasonable I mean I have no I have no problem with Hume but um yeah I think you're right constraint certainly gives us something else does how does constraint therefore fit into active inference because we talk a lot in active inference of hidden causes and latent causes um you know one or the other uh which implies some kind of underlying acceptance or Axiom that things cause other things to happen right and that's in kind of inherent in uh generative modeling or basian design that there's quality is again that could we just deflate that whole thing just by adopting a kind of humean skeptical attitude such that we can kind of keep these words but we recognize they're just labels for something that we can't necessarily pin down ontologically or should we actually be discussing constraints and does that give us anything over and above uh the latent cause hidden cause out there sure yeah so I think in terms of embracing the hean skepticism thing that's what I don't necessarily talk about completing those terms but one of the things I kind of tried to go into in the book it's just in an appendix because it became kind of a degression was like what would it mean to kind of accept because there is I think you know one of the things that i' gets criticized for a fair bit is this conflation between statistical and causal Concepts um so you know to take a a whole L of statistical relationships and say this is a causal graph and these are like causal relationships and these are real boundaries and all that kind of thing and I felt I feel like sometimes that criticism is actually a bit potentially unfair in the sense that you know there are plenty of philosophers who going back to him who have defended that kind of stat what you could call a statistical reduction of causation and know there's tons of work in like Cal inference where people have tried to look at how how those are related um and you know there's there are there are philosophical World Views in which the world is just a series of you know SpaceTime points that stand in certain statistical relationships to each other and that's all the causality is um so I think that would be a I think that's like an interesting direction to go with active inference um it's not the direction I would go but it is like there is whole lot that could be said about like a particular metaphysical causation that would be really compatible with the kind of active inference approach and make that distinction between the model and the territory if the territory was that way a lot smaller um yes yes exactly no that's absolutely right it's kind of it's nice because I don't agree with that but I think it's like an really interesting position that could be developed yeah well as you say I

think uh well it's it's a life to do metaphysics causation is a lifetimes project I think um so we need some philosophers who have done that since they've been sort of 11 I feel like it's one of those things where you know yeah you devote a whole career to the metaphysics of causation and then hopefully you apply it to something like active inference that's a really interesting point that if we got a concrete handle on what it means for something to be a latent cause and it's not just this um well in perhaps an instrumental conflation of uh statistical cause and and an ontological and genuine cause we could collapse somewhat the map territory distinction is there it's just the question comes to mind is there anything else that you can imagine if we got a sort of proper grip on or if we said okay this is not just instrumental it's it's a genuine Claim about what there is in the universe we would uh we would get closer to reducing the uh map territory distinction is there anything else I mean I sense agency is one of those things as well right um because we could claim that it's a kind of uh more like a map feature rather than a genuine territory feature if you're thinking about the the Deep metaphysics of Free Will is there anything else that kind of came up that you thought at least there's a question here of whether it's a it's a map or or or or whether it map belongs to the map or the territory just to I for me the agency belongs very much to the territory I do think it's a part of the territory that is kind of missing in the for enery principles map yeah yeah in terms of I think the biggest move would be to argue for a statistical iCal reduction of quation I so like that's not another thing but I think once you've argued for that then other aspects of the fry principle become more realist um what do you think I mean uh one thing that I think about a lot because it's kind of what I predominantly focus on and write about is the self um so some people take the notion of well it depends on kind of your underlying ontology but some people do take the the notion of a kind of I think I kind of think that cartesianism is having a bit of Revival um I'm not saying that I subscribe to that but other you know other people might and I think in some ways it's it's aided perhaps implicitly by a framework like active inference because it very much forefronts the the self- evidencing agent and I it came to mind and I was going to ask it when we were talking about interiority versus exteriority because um but you know the conversation moved on but it came to mind that are we always at risk of uh implicitly invoking aelf when that would be uh a misplaced um addition to the conversation right like is the distinction between myself and the rock well perhaps implicitly and intuitively I think it's well I'm I have a and The Rock doesn't now this is relying on very strong uh dualistic ideas of what what self is I wonder whether getting stronger on what it means for self- evidencing like is it purely a phenomenological claim that we have a sense of self is there something underneath the pure

statistical mechanics and the free energy minimization I just feel like that would get us a lot closer to a more realistic account have you been thinking about the self in any yeah so I think the way I'm thinking about the self the idea of self a self that distinguish itself that distinguishes itself um is very tied up with me for agency with agency so I think those two things are kind of are part of a package because the way I think about agencies in terms of autonomy in terms of giving a law to oneself all of that kind of um so and perhaps this is where I think when you take the frity principle as metaphysics and you take it you take something like self- evidencing as being what gives you the boundaries of the self I think what you end up there my impression of it is more like kind of a position that perhaps you would be more sympathetic to a position in which there are no real selves because everything can be kind of not everything but self- evidencing is kind of everywhere right um if if causation and inference are understood in this very very minimal way where all it is is to have the right kind of statistical relationships to the things around you then you going to have Mark of blankets all over the place you're going to have the ability to describe one thing as modeling another all over the place if that's all that you take the self to be then you're either it seems to me going to end up with a very psychist view of s or a view in which well if s are everywhere then there's no not really any such thing yeah I think that's reasonable I mean I I maybe I am lending myself towards pan sism because I do think that um phenomenal experience is sort of the one indubitable fact about the about the world so uh that's just that's that's kind of we can I mean I want you know we will get to ious I'm sure but without without going there yet and perhaps it will come up now is if I adopt if you don't adopt that kind of deflationary account of selfhood what is there in your ontology that's over and above that so we spoke about agency and how it was about sort of a law onto oneself and uh autonomy and that would okay there we kind of stake the claim about ontolog or you know on actually what's you know what might distinguish me moreck is there anything beyond Consciousness it might just be Consciousness that in a sense adds to this notion that I might have a self or might be a self whereas a rock isn't what is that over and above um this very deflationary account that I'm proposing or hypothetically proposing yeah I mean so I think one of the things you mentioned earlier is like the role in which constraints play in the for princi and the role that constraints play in pretty much like I'm I'm not a mathematician um so I'm sure but in pretty much every mathematical model you set up some higher level constraint yeah then you which is expressed in the form of invariant equation and then you can have you know lower order equations and parameters that change relative to that higher order equation but at some level you're always going to have that

stable higher order equation and that's basically a constraint like an equation you know the kind of in so I don't know if you've ever come across Robert Rosen at all I don't think so okay um so basically he was a I guess theoretical biologist um and had this description of living systems as systems which generate their own equations they generate their own functions nice um so you have two things essentially you've got your variables and your functions and your functions are for him God I'm terrible on my four causes your functions are your officing causes and your variables are your material causes so your variables are the stuff that goes in um your functions are the things that does the thing that does something with that stuff um he calls them efficient causes I think there's an argument they might be more like your formal cause your structure um whether they're formal or efficient is I think an interesting discussion um but the idea there is that what you have in the living system is a system where you don't have that higher order independent equation that sort of sits on top and doesn't change instead you have this cycle where every sort of function making up that system is modified by some other part of the system is that that that's okay that's an interest so well okay so that I feel like in some ways comes to this deflationary account of the self right there are just what verella called selfless selves without some superordinate homunculus that's that's guiding the show um okay that's but do you get what what what is what what's left in the concept of self if that's the case if everything is just precarious if the self is precarious in the that there are a multiplicity of selves that rely on uh one or the rest for the for its own existence can we be left with any cogent concept of a self at least from what the word we think means yeah I mean so I think you don't you definitely don't get like a cartesian self you don't get a lot of the stuff that people pack into the self I think you get you get for example an ability to be there a way of deciding if something is part of me or not a part of me um so that's what you get with the cell right the cell lets things in it doesn't let other things in and that doesn't mean that something is that is not a part of me or not a part of my cell can't be a part of a broader thing that we're both part of so like you know a cell in my finger is not a part of a cell in my eye right they're both part of a bigger self that's me um but with the like this kind of recursive uh what for some reason called the Mr closure but this kind of closure functions what you don't get is anything stable I think that's the really interesting thing that's different about this notion of self you get that kind of individuating network where everything is defined in relation to something else and some things are just not part of it because they don't enter either as a flow of energy into the network of constraints so functions of constraints um functions of your constraints and the variables changing or the input of

energy that does allow you to demarcate a sort of cycle that other things are clearly not part of because they not generated by it nor do they enter into it but what it doesn't give you is any particular thing that has to stay the same yeah yeah apart from what does the boundary have to stay the same not necess there has to be a boundary but there's no reason why the boundary couldn't change right so do we then say I mean this is a very sort of caterpillar butterfly problem or or you know so there are some great work you know I spoke to Mike Le uh about a month ago on this podcast and Andy Clark has got some really nice papers on this kind of process ontology perspective um are we then invoking uh like if we just take the lifespan as our sort of time constraint our time boundary are we invoking multiple selves then or a kind of superordinate system to which this metamorphosing um multiplicity belongs is it is it I mean this is going to come to this conversation about really like is that can we if we keep recursively building out a level bigger levels of abstraction and gen generality is there some underpinning rule is there some underpinning guide to that process ontology such that I can point at something in the universe over the span of a lifetime and say that was uh Kate or that was Darius or or or we just you know are we just actually just multip are we just multiple multiples you know I think sort of both yeah I think one thing that kind of annoys me with pris ontology sometimes is I think it's a bit too you know the selves all over the place this could be a self relative to some perspective but this could be a self relative to another perspective so what I like from the kind of closure view is you do get it if we're looking at this scale then this is the self and this isn't like that's not relative that's not quite as pluralistic as a the more pluralistic ends of the process ontology um but it's always going to be the case that if you take again the idea of constraints sorry was talk uh as you should um but what counts as a constraint is always defined relative to some other process so on sometime scale this is going to be an invariant thing that's not changing but channeling a change but is invariant itself but on another time scale that's going to be changing and going to be need to be replenished by some other activity um so those things are always defined in relation to other things what they're not defined in relation to is my interests as an [Music] observer so you do get this sense of yeah multiple levels of selfhood but you also get this sense that once you fixed a particular time scale what is going to be a self or a particular space what is going to be a self under that is not relative or variable yeah yeah um I'm just wondering whether if so Rosen is a biologist I'm wondering whether is there is there a limit to like over that time scale sure it's not uh it's not boundless there are limited forms that it can take it the constraints of that it sounds to me I mean the biologist will say this sounds to me like the actual physical body is very much one of

those things right um now I know people will say well the cells in your body change every however you know however often in in the course of a lifetime but um you know you say the boundary can change to what degree are we taking the the epip helium or wherever you want to carve your boundary to be a significant um self- evidencing if you want to use that term sort of limit yeah so I think the interesting the interesting it's what this I think in all of these things you have this really interesting tension between there's a sense in which it can change and there's a sense in which it can't change yeah yeah yeah but that at each moment I would say the current configuration of the system is pretty constraining on what it can do right so in each moment what can happen next in the very next instant is relatively limited yeah but how that Pro as with Evolution and you know how that can unfold over time is entirely unpredictable right because it can open up because these these changes aren't just I suppose quantitative it's because sorry which quantitative qualitative because it's not just say an increase in size it's not it's not just that the size can only go up by this much each time it's that once the size goes up to this much a whole other range of possibilities become available like so you get these discontinuous sorts of change yeah at point if you follow it through time the possibilities will be small over time they just expand um beyond measure so like there's this very nice notion that Stuart calman has is like the adjacent possible so this point maybe your possibility space is relatively tractable but because as you move into one area of It new areas open up you can't just constrain yourself to one area of possibility space in that way for a living system so I guess time yeah so the the the spectrum of time is really very important here in terms of the possibility space of constraints um I found it interesting you said it's not predictable because I understand that uh there are two ways you can kind of frame that one is is not predictable for me as an energy limited human being scientist uh and that's because I'm energy bound um and as you say once time gets beyond the $t+$ one it becomes combinator explosive uh as to the possible State spaces that I can be in okay that's one way of putting it the other way saying it's not predictable is that it's not predictable for ll's demon I it's not predictable uh in the sense that there are genuinely different options for like there is it's not the this the system really could have done otherwise right that's the classic phrase as you know in the sort metaphysical Free Will so which one of you kind of pointing to that is it genuinely the fact that because of the combinatorial explosivity of State spaces um it's unpredictable for me which is fine it's unpredictable for me what you're going to do next anyway so I mean okay that just shows my limits or is it genuinely the case that there is not there is more than one path that A system can go down and therefore even the best you know infinitely knowing mathematician would be

unable to write down the path integral that describes the uh you know movement over the time the DI chronicity of this being yeah so the second argument is something I'm starting to think about but I I think you just have to be so careful in terms of arguing that they are genuinely random events I think what you have with living systems is certainly the case that you can have events at a level small enough so they could be genuinely random can have a kind of effect that leads to a combinatorial explosion um and that's very distinct to lyricism so you know even with like you know you get um macroscale events from like microscale events even just in computers right which is really cool like you can have these single event effects where random particle emissions flip a bit or something and airplane crashes right which is super cool it's super cool that you can have that kind of macro scale causation from something as small as a photon right and that's just that's solely because of you know the incredibly small size of modern computer components but what you don't have is that having ongoing repercussions that's what you could have in living systems where something like that scale of event not only causes something to crash or something microscale to happen but then opens up a new region of possibility space then opens up a new region and like you get combination yeah I guess the question there would be I mean there are two questions there one is is it not the case that something like a human would have an attractor set to which is is almost uh again I'm not invoking any deep teos here but it's almost responsive to to those perturbations right that's kind of the classic vellian response is that all we have are perturbations and uh we have these I mean he probably wouldn't he didn't turn in this way because dynamical systems theory came after the kind of autois but we have this return to this attractor set such that I have this itinerant as Carl would say movement towards the attract set but that doesn't mean I'm always in that possibil like that high possibility that peak of the gaussian I can I can sort of be pushed away but I'm always returning up the hill in some way that's one question and then added to that at a more General level even if uh you know someone might come along and say sure you have these random Black Swan events and uh they have these trickle effects such that it's impossible to predict where and you know t plus 10 I'm going to end up because combinatorial explosivity someone might come along and say yeah but there's actually just a even bigger attractor set which governs the transition of you moving into this place which seems random but actually like the photon ing the plane it was always going to happen and actually it was in the plane's ontology that it had that attractor set of ending up in the bottom of the ocean yeah um so I think this comes back so the reason I haven't focused on the randomness aspect so much is precisely for the reasons that you've just been talking about is I think that's the less

important component the more important component for me is the sense in which I don't think I think the youve reach a point where to continue arguing that there has to be some sort of stable attractor for that can explain everything that I do it becomes so General that the selves which is the self the the selves kind of disappear under that because this is an attractor that this is a sort of description of invariant behavior that we all share so we all you know have to couple anabolic and catabolic reactions in the right way to keep our metabolism going that's a very general principle that we'll always preserve um but that doesn't give you any sort of individuation that doesn't give you a sense in which you're different from me and we're all different organisms so you can have very general invariants but those General invariants don't allow you to pick out individuals or you can have individuals but those individuals I think don't have invariance and the way that that as I understand it tends to be handled in the free energy principle is you kind of go to this sort of middle level so if you're looking at what's the invariant you know very simple example um if you take something like a EOL bacterium and it has that ability to switch its metabolic system you know from glucose to lactose and you can say well you know what's the stable model for that ecoli because up to this point it's only metabolized glucose and now all of a sudden it's doing something completely different that's a phase transition and Its Behavior and energy principal says it should die you just go oh well no that's okay because there's a higher level model for a colite in general at the level of the species and according to that higher level species model it's stable yeah exactly yeah and I think the Reas I think that doesn't work and I think the reason it doesn't work is because individuals can also persist through transformations in their behavior that are novel at the level of the species um so you know you can have you have the really cool examples of metabolic plasticity and cases where things like bacteria do like swap their DNA and gain new abilities and they might gain at the point of which that happens there's going to be an individual who gains an ability then no individual of that species has ever had before it's truly novel it can still be good for that individual and it's still part of that species presumably well yes we want to say we want to say it's the same individual yes and it's part of the same species but what's occurred has never happened before at any level of that individual's kind of generative model so is the so under this is the butter is the butterfly the same as the caterpillar yeah yeah yeah Mike said this really amazing thing which is that so butterflies I'm curious about what you're think about this butterflies and caterpillars eat different things right so caterpillars like leaves and butterflies like nectar but if you have so you can uh through sort of very low level conditioning you can sort of make a caterpillar go to a specific Leaf of a specific color because

you've trained it there's food there's going to be food there and what he said to me I haven't actually looked at this is that uh you it when it becomes a butterfly it it stat statistically goes back to that leaf more than by chance which implies that despite the and so the kind of big question there is where's the memory right if you're a neuroscientist you're saying well uh I don't know what the size of a caterpillar's hippocampus is but it's clearly it's completely changed right the architecture is completely different and yet the cognitive function or the output of that cognitive function seems to be the same um I'm not asking you to solve that Paradox don't worry uh uh but what I am wondering is does the free energy principle solution just to say well just broaden out the Markov blanket right and just say well we didn't know right we didn't know that the the eoli was going to become lactose metabolizing we didn't know that the caterpillar was going to become the butterfly does that make it does that reduce its import right like is there a problem this is kind of a meta question is there a problem with something that's just an hawk or post Hawk description um and if it is just that does it really I mean what what does the free energy principle say if it's just a post hoc or ad hoc uh description yeah um so I mean is there a problem with it it depends what you're doing um if you're trying to build an ontology of living systems then I think it is a problem um if you're trying to describe what living systems in your experience have tended to do in a like succinct and convenient way then it's not a problem at all um and I think you this is a thing that comes up when I make these arguments about how living systems can change in all of these completely not necessarily completely unpredictable but completely so there's kind of like a neologism that St is like unpre statable in the sense we can't write down which people hate and I but I I I it's useful it's useful you don't want to say predictable what I mean by unpre statable is we can't write down equations in advance for that individual so it might be that all of the equations of the entire universe would allow us to predict it but there's not one particular set one particular mathematical model for that individual that we can write down that won't change um God I completely forgot my third was the No it's it's was that oh uh it was it was just what oh well the real question was what does the free energy principle say like okay it's well I know what it says right I know it says that things look as if they are conducting variational basian inference over you know their internal States look like doing that over external States but really what philosophically speaking what does it say about things as they are in the universe if it's just this post hoc description of not only this not only like even if you have the path integral description of fre energy principle I'll give you both the state space and the path integral for formulation if it's all only saying okay I'm just describing how things uh

you know the attractive set that they tend towards and then also the path that they tend towards what does it actually say and I don't mean here in terms of utility right we're being philosophical I don't care about whether it can build robots or not I'm asking just philosophically does it actually tell us anything ontologically about what is going on in the world so I no I don't think the FR principal does to be honest um but I think it does raise like an interesting problem for the I think it's a utility kind of argument but I think the fact that it does seem to work quite well for allowing us to describe lots of things does raise an interesting problem for this view that if living systems don't have to stay stable if they can change in all of these if they can completely change their entire structuring configuration why don't they do it more often right because they do tend to be stable like right so when I make these arguments about that kind of unpre stability you can look at me and be like yes but we are homeostatic you know we do tend to stay very stable our body temperature is pretty consistent um and that's where what I've been really concerned with is the fact that that's not necessary it doesn't have to be the case that we have to be stable in that way to continue existing it's really interesting that we can identify something that continues to exist if it isn't stable so it's not a necessary principle on an individual system that makes it even more interesting to ask given that it's not necessary why is it so common um and I think the position I sort of end up at is you have this kind of system that can change in all these unpre stable ways that that would make it sound like the free energy principle isn't going to be useful for allowing you to describe that system clearly it is and I think it is because there is something right about the free energy principle in the sense we are trying to keep ourselves stable I think it's right that we're trying to do that a lot of the time that we are trying to resist new disruptions not always but that is a part of what we do and a part of the reason we do that is just this kind of suppose like cian gamble in the sense that this has worked for me in the past sorry hum not human will probably work for me in the future because that's the best we can do not because that's a necessary requirement like it's an approximate strategy to what we're really trying to do which is to keep this metabolic cycle going yes yeah there are lots of things that come up there I mean one is uh that I'm probably a little bit more skeptical about than you about the kind of uh imperative aspect of this um that humans are actually trying to do anything at all uh at least they might be but I don't think the free energy principle gives you that yeah I think the free energy principle gives you like if there is one if there is one human at this point it's the same human the next day what do they look as if they've done and again that's like hyper deflationary but that's genuinely all I think it does give you um you would need something like evolution by natural

selection or some like some more processed Theory um rather than the principle I think to give you what humans are doing right because as you say um we don't need to do this right like I don't like there's no there doesn't seem to be an prescribed law of the universe which says that I need to persist which makes it quite remarkable that I do and I think uh I haven't really thought about this enough because I'm not a physicist but I think this is beyond the pay grade of the free energy principle uh which is why things abide by they do they look as if well they do in some very inflation away but why they do I think there are some arguments I think Carlo relli has some arguments about why there are these neg entropy pockets in otherwise increasingly entr entropic Universe um I have to speak to him about this um but yeah it's it's fascinating I mean I mean again I won't go back to it because I think we've covered this a lot but I think the fact that you said that the SE the the self whatever that is can persist uh even if we disrupt this kind of attract attractor set idea like even if things don't tend back to equilibrium that the same thing can persist I think is a really interesting idea um I don't know how much I agree with it only because I don't know what if we're talking about the same thing when we talk about the self or the same thing or what equilibrium really is I completely agree that uh there is no such thing as perfect equilibrium right I mean I mean in of itself we need information transfer to act to for there to be anything right if not it's just well there's nothing right things need to be changing um what that means in terms of identity um I need to read the book I have to preface this by saying I haven't read the book and I will read the book but I haven't read the book um but I what I have read and what I want to go on to your more shorter although some are quite long uh philosophical papers uhuh uh and the the articles so I thought they were really cool really really um insightful uh and and yeah very very thought-provoking and you as you say you focused sort of before the book on perceptual phenomenology or or or what kind of more specifically visual experience and sort of probing this question of to what degree is perception enslaved by action right like perceptual experience as is as can be attested in the phenomenal ology uh why does it appear as it is and how much has that to do with action so I guess the first question I'll ask you very generally for the audience although I kind of have a notion because I've read the papers is to what degree is perception the slave of action uh and to what degree is it not yeah so I think my thinking on this is very strongly influenced by um philos called Susan Hurley who very early an activist um well she doesn't describe herself as an activist but she's I think meet I think she yeah I think she absolutely makes sense to be described as an activist in in quite a narrow use of that term so not just anti-representational um but as in being interested in the dependence of acceptional action and her

kind of big focus is on something that she calls the myth of the giving well sorry it's not her big that's one of her many focuses um but so the myth of the given is the you know this idea that we just receive conceptual content from the world in perception and a way of avoiding that is to go well no you know our experience is constructed relative to our actions but then you're presuming that the content of an action is already known that you have an account of what it is for an action to have a goal I think this move to say perception is for Action is sort of correct but it doesn't really solve anything because then what is action for um and too often people too often I feel people sort of stop at this perceptionist for action and look that allows us to explain a lot of things about perception but they're already they're kind of presuming a concept of action and what it is for an action to have a purpose isn't already that well understood so yes perception is for action action is also for perception you know it's a it's a circular dependence yes um but what is this whole circular dependence of action and perception itself in service of is I think the interesting question that kind of lies behind that well active inference gives you a nice answer right uh that it's well does it uh maybe predicted processing does actually yeah actually let me ask you does active inference I predictive processing I guess does because it has some underlying teleology that it's about prediction error minimization yeah case I mean you know it says how things seeming you know how the brain seemingly minimizes prediction error when you say uh why does why is uh why are perception and action coupled in this interesting way and you talk about sensory motor contingencies and whatever and the answer from active inference person would be like well it's all in the game of reducing variational free energy is that a satisfying answer for you no good good because I think you can you know you can say all sorts of things are in the service of something else but for that thing in itself to be a purpose process that's very difficult to make sense of right um and there's a really nice paper by I think it's Tony Cho and gosh I forgotten his first name after for Michael Anderson yeah there's a Michael Anderson y terrible this is the problem with only reading people's papers I know I know about it yeah and it's like really short but they just make this distinction between like Notions of predicting like two Notions of prediction so you can have one thing is predictive of something else a predictive of Rin um and if you just use prediction in that sense then you can say everything is in the service of minimizing prediction error because everything is in the service of being correlated with what it is in fact correlated with right so if something happens the thing that is most likely to happen after it is particularly likely to happen it's incredibly deflationary boring incredibly deflationary so you can have that notion of prediction and everything is being prediction

minimizing that does allow you to describe all sorts of things as prediction minimizing which arguably what the free energy Principle as just as an addendum arguably the limit of the free energy principle um yeah so I think if all you're trying to describe a predict what you can call predictive relationships all very well and good but a predictive relationship is not a prediction yeah it's not a predictor um so to say that everything is in the service of minimizing free energy or minimizing prediction error there needs to be an account of what makes the minimization of prediction error a purposive process as opposed to just I look at these two things as predictive of one another and see that they're correlated what makes it an attempt or you know what makes it a goal directed prediction that's what's not in the free energy principle it's not what's that's what's not in prediction I've processing there we either go deflationary or we presume that there's an agential and predict predictor a predicting process going on it's not really predictive um but what makes it a purposeful prediction is exactly what isn't in this framework I think yeah okay cool yeah I mean arguably I mean I'm not going to I actually have no bone to pick here with any of this I'm happy to leave it all open but arguably it's because there isn't any purposiveness or agency but anyway you actually took my question way deeper than I was going to than what I meant by it which was that um but I like that it's a really it's a nice add on to the conversation we've actually just been having um I meant more that uh you write and I think a lot of this is influenced also by Andy and Andy's work and and it's almost a response to Andy's work which is that perception in many ways um is nothing more than the process that kind of puts you in contact with action by that I mean that uh we perceive the world in the certain way because it licenses us to act on the world and I've been kind of thinking this almost in terms of Consciousness more generally maybe what Consciousness is is kind of course graining filter that permit action upon those phenomenal contents and here you talk about um the intermediate level representations or intermediate level processing so what did you what what what is intermediate level processing what are the kind of claims of that uh position that theory yeah I think so I think trying to remember who actually I think it's Jesse Prince and Frank Jackal for they yeah they're the kind of original intermediate level people um and what really nice about predictive processing is obviously well anything hierarchical um is that you don't have to have just one intermediate level there's levels you can move up and down them um what the intermediate level idea gets at is this idea that what we as agents act upon and not you know individual sub subatomic particles um yeah you know they're not we don't act at that level and we don't act at the level of the planet you know we act in a level of you know glasses and bottles and C stuff um and I think that's

nice but what I became interested in in through processing was that you know we don't act on one level of real of reality or SP temporal scale you know what we're interested in acting on depends on the circumstance and there's lots of nice sort of perceptual psychology experiments but shows the level of detail we perceive also depends on the circumstance um of what our actions are relative to so there are these things called naon figures which I don't I'd be nice to be able to get picture one up actually but basically they're just a shape made up of lots of little smaller shapes yeah yeah yeah yeah you can have like a h made up of little hes right um and there's a general Global precedence effect so people tend to see the bigger letter rather than the letters that make it up unless you present that in the context of asking someone to read a paper right if they're asked to read a paper then they'll they're in reading mode and they're perceiving at the level of the individual letters if you're asking them to identify a figure they're perceiving at the level of the hole um and you get nice effects where this doesn't just happen on a sort of immediate situation basis but based on the kind of actions that you habitually engage in so one of my favorite results and I cannot remember the citation right now it's I think it's Lee the main author her name is Lee um but it's a paper showing that people who play a lot of Call of Duty of Duty one I like that one dramatically changes I really love that result yeah so people's yes we're sort of perceiving at the rough medium level of the hierarchy but that's going to vary a lot with the kind of things that you do and you know like you can imagine a visual artist's perception is going to be very different for that reason because they are operating perhaps at a finer grain or even perhaps and more cor scin with a good degree of regularity um yep so yeah it was kind of taking the intermediate level view of Consciousness and saying if we've got more than one middle level like is it you know how does that actually shift with action and it makes nice sense of lots of experiments like some sort of intentional blindness type experiments and that kind of thing really nice yeah so this is I can just tell people this is Lee and colleagues 2009 enhancing the contrast sensitivity function through action video game training Nature Neuroscience yeah really cool I mean I I have to take this opportunity to plug myself um funny enough because before we even started speaking it's it's weird I went on this I just wrote this paper called um it's titled currently how preferences enslave attention uh calling into doubt the exogenous endogenous distinction from the perspective of active inference and it's really funny because I try and give a really formal account of exactly this which is I think that um attention is ultimately governed by the prior that we possess and there's a really but I think what's interesting about this is um the interplay of the preferences mhm so for example I I talk about this in the paper I say um why do I uh you know why am I paying attention to

speaking to you right now but then a fluffy dog might walk into the room and suddenly my attention is all geared towards that but then I have this sort of counterveiling preference to go back to speaking to you and then suddenly if the building starts burning I'm going to have absolutely no uh choice but to pay attention to the fire and and to jump out accordingly so it seems like in addition to the high IAL generative model that Maps quite nicely onto a hierarchy of attention and I'm trying I've been trying to uh answer or explain what governs that hierarchy and I'm just curious about I mean I can send you the paper um because it really is very very very I mean amazingly convergent with your paper and I should have read it before and it's my fault but you are now cited so don't worry um because I've been reading obviously your work I've been thinking about so for example why is it that when I'm completely ravenous all I can pay attention to is the fridge and you know the the food around me like the sources of food the potential sources of food and I argue that I think it might be twofold one is that uh those preferences are encoded deeper in the generative hierarchy than other preferences and I think that the reason why that's important because if you're a predictive processing person that means that the predictions issuing from that deeper level are modulating all of the more superficial levels so I think that I think that stands right because it's like what if there was something above it that it the thing that we're talking about would itself be modulated but my Hunger is never really modulated right like if I'm starving there's not much more that can top down change that so that's one thing the depth the other thing was Precision waiting I think that these preferences right so why when I'm ravenous uh do I go out and get food um and pay attention to food more than anything is because that preference gets extra precision waiting extrinsic gain um I'm curious about just generally what you think about that proposal and whether there's a like there are added Dimensions to this kind of um attentional hierarchy that need to be countenanced I think what I like about it what you how you've just described it in particular is when you think of something like hunger and the kind of significant role it plays it's just one of those nice cases where you have this if you know if you have if you approach this view of like a hierarchical you of cognition from a very maybe classical Cai perspective your view of what's going to be at the top what's going to be the really fundamental thing that's modulating everything else it's so different from when you approach it through actually you know being a human being in the world who gets hungry yeah realizes that you know the things that modulate everything else you do aren't things like the truths of logic right right um so yeah I think I like that idea that I think hunger is a really good example of where you see that you know your hyper priors or your fundamental priors

aren't is purely rational pure like sort of rational detached truths about how the world Works they things about their needs essentially you know that's what's at the root of it all is needs um and how that relates to all the like attentional switches we actually make in everyday life you know when we're distracted by like a cute animal or like how that emerges out of these really fundamental needs of obviously super hard to draw draw out because it's so so complicated but yeah when you have something like hungry you do see Hunger makes it apparent well because there seem to me to be a uh there's a dynamic element to it so I think uh what I say also very explicit in the paper is that we're not always paying attention to being hungry because if I did I would be obese right like I it would just be really really bad for me um however that so that that's one dimension the other dimension is the that however when I am hungry it takes precedence yeah there's a kind of it's not a paradox but there's a kind of tension there which is that it doesn't always seem to be occupying super high Precision weight yeah but when it when I have prediction error or free energy with respect to that preference there seems to be a prioritization of the fulfilling of that preference so I I I invoke something like a higher order AC mental action policy which is something like when there is free energy at this level put extra Precision synaptic gain on that preference another interesting idea I could run through you know could run through with you is the idea that I don't think that I think these priors these hyper priors or phenotypic prior whatever you want to call them they might have inherently quite High Precision weight and I'm curious about what you think of this just phenomenologically is it the case I've been trying to think whether it's the case that um in some ways a very phenomenal experience is shaped by these priors in very subtle ways and I think that comes out in the fact that um I may not be aroused or hungry or scared but if there's a loud noise I'm immediately paying attention to it or I might notice uh you know like there's a reason why advertising works right I might find myself getting drawn to a poster of a burger or whatever which implies to me that either this this higher order policy is very sensitive very contextually sensitive and at the slightest contextual cue it will just deploy High Precision or there's a kind of standard Baseline level of precision weight that these preferences have such that when they reach over it's very easy for them to reach over a threshold such that they grab attention I'm not quite sure how to unpickle yeah like sort of a low threshold for Activation rather than yeah so I'm not I'm just curious about what you think about I mean I'm curious really fundamentally about what you think about the phenomenological idea that our experience is permeated by those preferences right our attention in many ways is permeated by those preferences is that the case or can you imagine sort of is it the case that if I am well sated

and so on Hunger as a or or satiation just basically doesn't affect affect my cognition whatsoever what is there a kind of what's the kind of blurry level of influence do you think that those preferences have yeah I that's a really interesting point like to what extent is your phenomenology shaped by your needs when your needs are satisfied right right um I don't know I'm not sure how to think I think that's I think it's an interesting question I haven't thought about the V visual phenomenology stuff in so long whenever anyone asks me a question about it there's that tendency to sort of go and be like what is my visual phenomenology um I mean I mean if anyone's listening if anyone knows whether Merlo Ponte or whoever I'm sure someone spoken about this um so I think maybe one aspect is I suppose maybe what I would say is I think that we those are like going to be you know your basic fic needs but I think as you become a human being as you become a person this is my M Pony is not great but I know he talks about some stuff as you become a person you develop things that take on the same kind of status as these more basic fpic needs yeah I suppose what I would say is I don't think there's ever a situation where all of your needs are satiated yeah because you all you develop other needs that you know they don't initially take on the level of a need so like you know a desire to have your parent maybe desire to have your parents approval is probably right probably want deep but you can you can pick up you know new things that initially aren't need but they become so integral to your identity that they become need and then those needs are shaping your perception in a way that when hunger isn't shaping your perception that's not to say that no needs are it's just to say that a different need takes salience um and I think this is a really nice this idea that's what beautiful to me about this idea of need as being fundamental need and ident like constitutive needs it's that it's very amable to being found both at the biological level and at the social level so there's a philosophical Christine cosgard who has a little of work on this idea of self-constitution that as humans we constitute our identities and they that gives us needs that we have to act upon to constitute ourselves and it's very Canan but it's very sort of rationalist I mean she's interested in animals but is generally described in quite rationalistic human conscious terms but it's the same kind of thing going on there as when you're hungry and you need to eat when I'm not um acting in a way that's consistent with with my my own view of my identity there's like a need there that I'm trying to fulfill in failing right the same way as when I'm hungry and I'm not able to get food there's a need there and I'm trying to fill in failing yeah I mean I guess that too far off Bas but no no no I think that's absolutely right I think just thinking about it you know even now I think there's what I I wonder whether there's always a sort of trickle back to some overarching hyper prior right so

uh I want to write a great paper because I want to impress my parents well it's kind of Aristotelian in it way well why do I want to please my parents well because I want I don't know I mean why would one want to e so they don't I don't know evolutionarily so they don't throw me in the bin or whatever um and so on and so forth and I was just wonder whether that gets where that gets bottomed out but I think that's also what I mean by this idea that these Hypo priors permeate our lived experience because even if this is actually just an instrumental description I wouldn't say all of all Behavior or all perception can be traced back to a limited set of hyper prior but I think some uh some can right so some very superficial goals or seemingly superficial goals like finishing a paper yeah have the capacity to be really embedded within a deeper preference of being happy right ultimately I guess the only question there would be h hunger something like hunger or more phenotypic needs they almost seem less like that because they don't seem to engender this kind of deep hierarchy right there don't seem to be plenty of things I do because they indirectly contribute to my so Society um they almost seem like a different type of Prior such that they need to be fulfilled but they're almost singular in the fact that fulfillment can only come through like particular action policies anyway this is super off I'm just sort of but I guess they're interesting because they're both very basic and very specific yeah like hunger as a need is very fundamental but also there are very specific things that can fill it it's not very general whereas if you take a more basic need like I need to have a coherent identity I need to have a coherent personal that's everywhere right that's just everywhere yeah and there's so many different ways I could fulfill that so you might be able to trace back my having this conversation with you about my philosophical work work to a need to have a coherent practical identity but you can't get from the need to have a coherent practical identity to this particular manifestation of it right it's very underspecified um there's a really nice I think it's it's idea of like the the deductive importance of this kind of idea of an identity so it constrains what you can do but in such a limited way that there's so many different possibilities that they're compatible with and what's interesting about hunger and these more phenotypic needs like I think is that they are really fundamental but also a lot more specific you know you can derive what I must do to fulfill it um so I think that would be a difference yeah no that's interesting ah this going to be fun editing um that's all right that's what academic Academia for cool well there's there's I think just on that point another thing I think is interesting as well is when you think of I think you get this view where some of our needs are really manifestation of social leads so for a society to work it needs to be constructed you could argue it needs to be made out of individuals that have coherent practical

identities like if we don't have coherent identities if what I'm doing now is completely and utterly unrelated to what I'll do tomorrow then you have no ability to interact with me meaningfully right right because I can't make any commitments to you that's really yeah that's really interesting actually there's a sense there in which you can see why the need for my need for coherent practical identity so by which I mean just that you know being a sort of predictable person over time who's committed to certain projects and follows certain narratives if I don't have that then that's not just necessary for me but that's necessary for us as a whole and it becomes a very primitive need of a group of people which I think is interesting do you think that says anything interesting about sort of um it comes back to constraints in many ways yeah that um do you feel like there's one or or you know we think perhaps of the kind of more like the the hyper you know Aristotle would thought of a hyperv if he had that language as um udonia right and in the case of humans it would be it would be reason exercising our reason and and the way that you say well having a fixed Identity or having a set Identity or stable identity is Downstream on me being a functioning member of society and then if you ask the Marxist it would be because that affords you the opportunity to you know uh be a be a capitalist slave or something I don't know um but you know they might say Well in that way you're just a kind of your preferences are just a vehicle for a broader political sociopolitical structures preferences and that might be ground and then you know what's the authority over that well is there is there is there a point I mean I guess my question is there a point where those preferences ground out or is it philosophically vague as to how recursively you know recursive that can go yeah and I think so for me I think where those preferences ground out they in terms of scale you can have them being instantiated at lots of different scales that can be the preference of a society and the preference of individuals within that Society but what they ground out in is this idea of the ability to coherently constitute yourself over time because in as far as your identity isn't something that you can take as a given it's not just that you know like a rock you'll just continue to exist whether or not you do anything you have to continuously do something to you have to create yourself into the future in that way whether I exist in the future depends on Me My Success now um the con the ultimate constraints are going to be there are things that I can't do you know there are things that if I do them now I won't continue to exist in the future and there are things that I have to do to continue to exist in the future but in the space of things that I have to do to continue existing the possibilities are pretty vast um there's tons and tons of different ways I could continue my existence so from the fact that I have to continually construct this continuous self process that tells you very little

about how I have to do it it tells you things that I can't do but it gives you very little Traction in pinning down what I will do at each moment in time so in as far as there are these kind of ultimate constraints they don't get you super far in pinning constraining any particular individual yeah yeah yeah yeah well there's always that F you know there is always that individual versus Collective or that that that that dialectic right like and I guess that is that's imminent in an active inference more fundamentally right like unless yeah I don't think any of us have this kind of notion of a process ontology where it's totally loose that you could be anything um yeah I don't want to go back to that because we've I don't think there's an answer um talking about grounding out and bottoming out I want to go back if I may to this notion of intermediate level processing so so perception if I've you know read these papers correctly perception in many ways is is sort of the guide of action because it allows you in phenomenal space that which is emerges in phenomenal space is that which you can act on let's say right so it kind of very JJ Gibson idea um but the reason why you know I don't perceive the uh the atoms of my phone is because I can't grasp ATS right so there's this nice Mutual coupling and that runs down all the way down to the phenomenology but as you say very nicely in your paper if I take a brick wall I think this is a celian idea if I have a sort of brick wall there isn't one level of which I can perceive that brick wall uh I can stand away from it such that it becomes a kind of uh ambiguous mass of brick or something but I can't see the individual bricks uh or I can go even closer and you know this is a illusion of the cigar or whatever and I can I can Noti I can then lookig and there's kind of this fractal nature of perception so we're talking about grounding out and boming out is there you you speak about this a little bit in your paper but I'm curious about what you think now is there a point at which perception bottoms out um and can is that again just mappable to to action such that it grounds out at the point where I can't act right I can always grab the cigar I can always I don't know what I could do about with the sort of brick wall but in principle I could do something with it is there a yeah where does that where does that chain stop yeah I will I just need to get some water sorry yeah yeah yeah no worri sorry about that no wor so yeah what is perception ground at what level um I mean I know there's experiments you know you can look at all the sort of stuff on what is the actual capability of the sensors that we have um in terms of people's ability to kind of discriminate the presence or absence of a single Photon and that kind of thing um but in terms of phenomenology I don't know I don't know whether I'm trying to remember what I said in the paper because it's been would you like to know I would like to know was like six years um while I'm looking you can you can try you can try and remember um but no you do you do

mention it that I mean it it's I think I think it's totally reasonable to say I I don't think there's anything intuitively wrong with a notion that it will bottom out at the point of uh attend you know actionable inaccessibility if that makes sense right like it just doesn't make sense given the body that I have and again it's very sort of Merleau-Ponty idea that it just doesn't make sense to perceive atoms with a hand that is uh manipulable only kind of quite coarse levels right like I can't grip just because of the nature of the hand that I have I can't grip um atoms which makes me think it's a different question there's a uh I think again intuitively we think there's a kind of universality of perception because we're our sort of folk science or folk philosophy is telling us that we just receive the world and we recapitulate it perfectly a kind of direct realist um way of viewing things um to what degree does your argument that perception is kind of affording action in the sense that it gives us this intermediate level perception that's grippable and whatever to what degree is that kind of a sa WF uh account of vision I mean by that I mean to what degree I mean how much does affordance and action relativize and permeate perception is there anything beyond it um that that's kind of what I I mean I don't think any you know I really J I don't know I might get shout that I'm not sure Gibson really tells us yeah so I I think for me I would say no in the sense in or in the perception of objects the perception of an object I think is always cashed out in terms of action possibilities any time you perceive anything as an object you perceive something that you can pick up that you can rotate that has other sides to it what I'm I think the reason I'm not entirely sure whether that's exhaustive of perception as a whole is in terms of kind of sort of Altered States Of Consciousness type experience where you have perceptions that aren't really perceptions of objects anymore um you know so when you have I don't know whether it's some of the extreme states that people apparently get from meditation or from LSD and you have that kind of very non-ordinary visual phenomenology I think there it's very hard to see what the action is um behind those kind of experiences I mean the closest I've had to one um is playing around with you that Lucille like thing the dream machine thing and you get those kind of you know swirling mandalas and all that kind of thing yeah yeah yeah um and it's not really clear what the ACT it's not clear what the action is there that's very nice that's a good idea there is also the sense at least though with those experiences right like you know how they'll change if you move your your I think you do in my experience you have an awareness of how you know if I move my eyes this way the I'm going to get that sense the color shifting across my visual field this way there's still some kind of contingencies to them what there isn't obviously is any sort of sense of any kind of needs of like why should I act this way or that way you know I can

actually change it but there's no relevance of that kind of visual phenomenology to anything I might need or want in the same way that there is with like tables chairs lamps and water bottles yeah yeah yeah yeah that's that's a really that's a great paper idea by the way for anyone listening You' just got given a great paper idea to what degree what are the affordances in psychedelics I mean again I haven't thought about visual phenomenology in ages and it's it's fun to think about yeah no it's cool um I mean it makes me think you know we always think of affordances in terms of physical actions but the nice bit of active inference is people like lar SED Smith and others is uh the alignment let's say of mental actions and physical actions such that they can be described via the same formalisms and I wonder whether just really FR this you know it's in the dark I haven't thought about it whether the affordances there might be mental affordances in psychedelic States especially at this very deep level of let's say like a DMT trip or non-dual awareness um but again that said it's still that still foregrounds the notion of needs and uh a kind of autonomous being who has needs onto themselves because that is only the that's the kind of the only way that affordances make sense so you know I was going to ask you there you you say in one of your this I think this will be my final question because it's a big one well it's not actually a big one but it's um you know about it's comes back to the self I'm always going to come back to the self in many ways because again that's what I've been writing about and that's what I'm most interested in um you say you don't really go into this idea but I think it's really important that you say and I wonder what you talk about in the book as well you say in um your 2022 paper expecting some action having a temporarily fit counterfactually deep self model seems an obvious prerequisite for the presence of high all to self-consciousness but it is not immediately clear why it would be necessary for the emergence of more basic phenomenal properties such as the immediate visual experience of an apple on the table and then I think you go on to talk about um the this kind of action based account of visual perception that does give you the visual experience of an apple on table but I wanted to sort of when I first read that I was thinking yeah it doesn't give you it certainly an apple on a table but it certainly doesn't also give you everything that we can talk about in terms of selfhood so I've been very interested in uh what the phenomenologist called pre-reflective self-awareness um or what people also refer to as the sense of self as subject so this is the kind of you can you can be very deflation about it and just say um the subjectivity that's imminent in all conscious experience or the perspectival or the very sort of non-conceptual possessiveness that seems to accompany all or not even accompany it embed in all conscious experience and here I'm thinking about people like Sean

Gallagher and Dan Sarvey and all about to Hle and M. Ponte and I'm really really fascinated than them but I'm wondering whether you thought about that as well in terms of what act of inference can say about the self um there's a nice paper by Matas Wasak who talks about the I versus me and the fact that act of inference in principle can give you very much the uh the me the self as OB what I take is the self the sense of the self as object but it can't really give you the why which is the sense of self as subject um I'm wondering whether you thought about that at all in terms of uh Act of inference and the phenomenology I know you focus more on visual phenomenology but underlying all visual phenomenology is this kind of subjectivity this fact that there is a for means of visual perception so I'm just wondering whether you'd given that any any consideration because you did make that comment in in that paper I I don't know the paper that you just mentioned I'd be interested to so I don't I'm talking about so you made the comment about the fact that WN so yes so Wnac is I think it was in Frontiers um I can just for the AUD I and me the oh it's not is it I and me the self in the context of Consciousness Frontiers in Psychology 2018 I can send that to you yeah it's interesting um so sorry the question was more was on well the question really feeds off this idea that you said that the this notion of deep uh temporal counterfactual uh planning in many ways affords a kind of uh narrative self a kind of Rich self model that's capable of being reflected on right like I am a thing that causes things in the world this idea that we are a hidden cause we are the best guests of uh multimodal sensory information and so on and all these things that George is writing about and my response to George has always been well yeah that's very good for the self's object and by that I mean the self that can be reflected on so I am Darius I'm 25 I'm from London and so on yeah like that implies I'm a rifed thing that uh it can be cast as a latent cause and I think active inference gives you that what I'm asking is you said well it's not enough to give you this kind of very basil visual perception of course I mean I think that's kind of clear what I'm also saying is it it doesn't just give not give you uh visual perception per se it also doesn't give you the subjectivity or the very non-conceptual sense of self imminent in visual perception okay yeah so I'm just curious about what you think about that yeah and I think that I think that's maybe tied to this way in which I think the sense of self the the way I think of a self being very different from the way active inference necessarily does which is a sense of sort of a kind of intrinsic striving I suppose in the sense that it's not just I have this model and if something happens I need to respond to get rid of disruption it's that I am and also maybe to what we said earlier about this idea I think that there's never a position when all of your needs are satiated is always some sort of need it's just which one is being prioritized at

which point in time so the basic sense of self I think that is underpinning visual perception is going to be that sense of imbalance or misalignment super nice I about that yeah I mean that that again is kind of quite punching I guess but like um the B the basic self is like misalignment Mis achievement that we're trying to align and that's always a futile process because you can never get rid of all of this mis achievement you can just get rid of one element of it and something else becomes more Salient and then the kind of reflective self is what allows you perhaps to you know that's when you get into planning about it and all of that sort of stuff and perhaps in order to plan you need to act as though your needs are going to be the same so even though time in ways that you know you might there's no necessity behind your needs staying the same but if you want to make plans in the world you're going to have to act as if they will in order to project yourself into the future um oh that's a good question so a good question as to whether you can fake alternate desires that's a different question yeah that is really nice I mean I'm curious whether you think there's kind of um this nonreflective uh precarious self-awareness is do you think that man what does that manifest as in your IDE in this idea is it conscious is sorry is it um affect or is it even deeper than affect yeah so for me it's what it is it effect it's it's probably effect whether whether it's Consciousness I think I'm undecided on whether you need some sort of reflective capacity for it to be conscious so I would say you know every living system has this like basic striving and need a bacterium has it anything that is a metabolically a metabolic cycle has that um which if you say that that as soon as you have that you have you know something like Consciousness you end up with a very general notion of it um and I haven't really got to what I think I would be willing to I think it's less crazy to me um if you think that that's what it is is to be to have purposiveness and intentionality then you might be ascribed and you think that that's what is to have Consciousness you might be ascribing it to very very simple life forms the main thing for me is that you're not ascribing it to everything that that very minimal notion of striving is very general and weak that would be okay for me for that to be something like some sort of minimal phenomenality yeah yeah yeah yeah just the main thing I think is the most important thing to avoid if we want concept if they we want there to be to be realists about the mind about phenomenality is I feel that once that becomes a property of everything it becomes a less important or real property so even if it's watered down to the extent that we can say that there's something that bacteria have effective States that's still okay as long as um and tables don't Right From perspective yes well rather you than me to draw that dividing line um but I'm sure you I'm sure one can I mean I think just to sort of say what I'm thinking currently I don't think that this form of Consciousness needs to

be the product of any meta reflection there are people that think that and it's a very sort of these are not phenomenologists per se these are something more like um analytical philosophy's mind so people like Armstrong like and Rosen far I've been thinking about this because they argue that sort of for this this intransitive Consciousness can only come about because it's the object of a higher order mental state yeah but then I mean this is such a because it's obviously leads to an infinite regress which is like is that mental state conscious and if so how is it conscious without itself being the object and so on so I I don't know I'm I'm sympathetic again I'm not going to stake any solid claims I'm sympathetic to the idea that there is just minimal phenomenality that doesn't need uh that doesn't rest on any higher order cognitive function MH and you know is the fact that there is this uh at least plausible interiority about living things that autonomous or adaptive things does that feed into the minimal phenomenality is that the ground of minimal phenomenality such that it's kind of darine and it's this kind of imminent anxiety haiger might have even said you know that this future striving that is consciousness I don't know but I don't hate it actually um there are so many people who are trying to get to the BM of this what is this minimal phenomenality whether it's affect or uh whether it's salience or whether it's time or whatever that um it's hard to say but that that is a lot to there a lot to chew on I'm also aware of the time yeah I want to say just in terms of the reflection thing if that's what you're thinking about and you're not convinced by There's A really lovely book that I just came across recently called on reflection um I again first named um but it's corn BL I want to say Hillary corn BL and it's literally going through how philosophers have relied upon reflection the idea of reflection to get themselves out all of these problems with their first order theories we can't explain it in terms of this but if you reflect on it yeah yeah yeah then you have a self or then you have this and he's like you know if you haven't got the first sort of thing that you're reflecting on sorted then just moving to reflection is just passing the buck yeah of course it's a nice short book but he I think it is a really common move to sort of rely upon some sort of higher order reflection to explain how something you can't explain happens right um so yeah I I think that's maybe why I'm more interested in minimal phenomenality because I think there's no point talking about everything else until you figured out what that thing is and then you need can sort of explain how you can have reflection on it yeah I'd really recommend it yeah that's that's cool I think yeah that is cool I think the idea they would have is that I mean just in terms of this very sort of fact this idea that minimal phenomenality is itself Downstream on reflection on minimal phenomenality from some other mental state is that that other mental state that higher older

mental state is not conscious MH but then you just have this issue of how is a non-conscious mental state intending a conscious mental state anyway this for another day Kate this was really fun I mean a complete uh mind frazz and brain roller coaster but really amazing and insightful so I'm extremely grateful genuinely um I say it to everyone I I hope it doesn't end up seeming deflationary or or boring or as long as you get their names right when you say it that's the main thing I do try and get people's names right I do try and get people's names right I'm always like scrolling away to make sure I get the paper names right and whatnot but um genuinely so much to think about not only I'm sure in terms of my own uh case because I have now a lot to think about but hopefully in terms of the audience as well um I always like to give people just a little bit at the end just talk about where they can be found on the internet presumably rather than physical location because that would be weird um but yeah in terms of Twitter or email or whatever uh and also what kind of work they have upcoming that people can look forward to cool yeah and thanks so much for having me this is the first time I've done any sort of extended I used to be the I used to be a journalist so I used to be the person interviewing people so it's been really interesting being on the other end of the interview oh well you you smashed it honestly that it was nice it was a nice like chill first interview um yeah so I'm on Twitter I'm trying not to be as much but I am on there just as Katherine Nave um just all one word in terms of upcoming work there's the book coming out hopefully at the end of this year I'm thinking it's still it should be maybe by October but it could be February it's been in the works for a long time how long is the book may I ask oh I not sure I think I want to say like I don't know just just to warn people I guess I think not yeah maybe like 80,000 words yeah I think it's about 70 or 80,000 words sounds like a lot yeah but there's an appendix yeah yeah yeah and it's funny it's I've been told it's funny and this was built off your PhD you said yeah okay nice yeah um yeah it's supposed to be relatively accessible I would say like certainly you can you can dive into it and get what you want out of it so like the first half is probably I think is like a you know you think is a relatively decent introduction to the kind of basic philosophy of the fry principle and then the second half is more if you're interested in stuff like purposiveness and intentionality and an active concept you know more oriented towards that um there's also some blog posts I've got a post on dialectical systems which kind of summarizes some of the arguments of the book um and I've got a little substack post I don't have a substack I just have a single post that kind of tries to give an overview of the book's main arguments so those might be worth checking out um and the next thing I'm working on is a big paper about Hegel so that'll be a while but yeah well thank you again so

much from me um and yeah and and from The Institute as well I'm sure everyone's going to love it and and everyone should go get the book when it's out of course yeah thanks for inviting me I really enjoyed talking to you about it all right no absolute pleasure all right thank you everyone for watching as well byebye by